

WEEKLY PROGRESS REPORT

Week 2 – 07.06.2026

Project Title: Implicit Domain Adaptation Through Pretrained Feature Reuse: An Empirical Study of Lightweight Transfer Learning Across Heterogeneous Medical Imaging Sources

Team Members: Abhay Sivakumar Nair (220905199) | Shruthi Andra (230905223)

Department: Computer Science & Engineering, MIT Manipal, MAHE

Guide: Dr. Tyson B D Cunha

Report Period: Week 2 | June 1 - June 7, 2026

Specialization: Minor Specialization in Computational Intelligence

1. Project Abstract

This project investigates whether lightweight ImageNet-pretrained convolutional neural networks (EfficientNetB0, MobileNetV2, ResNet50) provide implicit domain adaptation across heterogeneous medical imaging sources, eliminating the need for computationally expensive explicit alignment mechanisms such as adversarial domain adaptation or GAN-based style transfer.

The core hypothesis, termed implicit domain adaptation through pretrained feature reuse, is that pretrained architectures encode sufficiently domain-invariant feature representations to bridge scanner-induced and population-induced distribution shifts. This hypothesis is directly motivated by prior empirical work in structural crack detection, where EfficientNetB0 outperformed a custom CNN by approximately 10 percentage points on pooled multi-source data with characteristic convergence instability in the custom model.

The project validates this hypothesis across three medical imaging tasks: skin lesion classification (ISIC 2019, HAM10000), chest X-ray pathology detection (CheXpert, VinDr-CXR, NIH ChestX-ray14), and diabetic retinopathy grading (EyePACS, APTOS 2019, Messidor-2). Domain gap is quantified using Maximum Mean Discrepancy (MMD) and mechanistic interpretability is provided through Grad-CAM analysis.

2. Work Completed This Week

2.1 EfficientNetB0 Phase 2 – Full Fine-Tuning (Backbone Unfrozen)

Building on Week 1's Phase 1 results (frozen backbone, best val accuracy 65.56%), Phase 2 training was conducted by unfreezing the full EfficientNetB0 backbone and fine-tuning at a reduced learning rate of $1e-5$. Due to Kaggle GPU quota constraints (30 hours/week free tier), training was conducted across three continuation runs, each loading the best saved weights from the prior run.

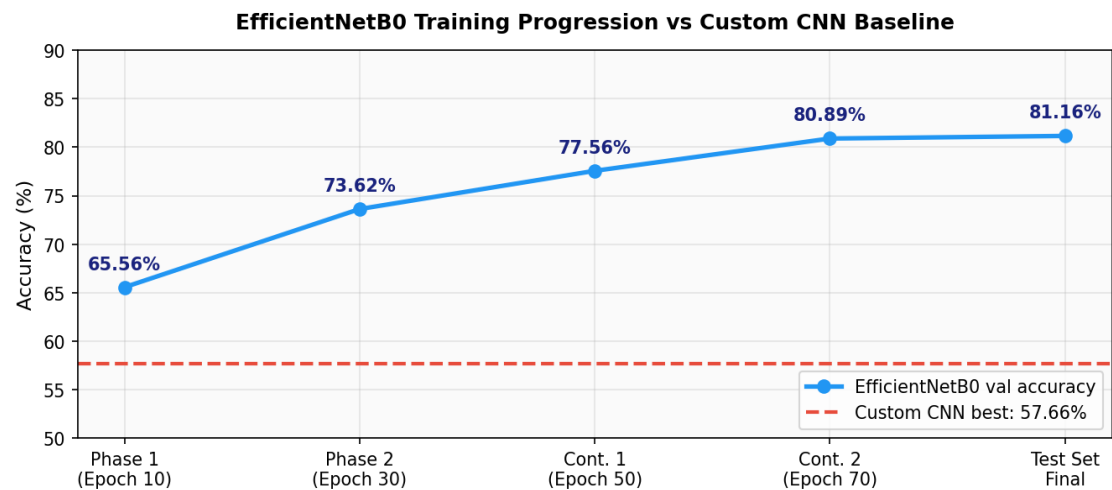
2.2 Training Progression

The table below summarizes EfficientNetB0 val accuracy across all training phases from Week 1 and Week 2:

Phase	Epochs	Best Val Accuracy	Training Stability
Phase 1 (frozen backbone)	10	65.56%	Smooth, monotonic
Phase 2 (backbone unfrozen)	30	73.62%	Smooth, monotonic
Continuation 1	+20 (total 50)	77.56%	Smooth, monotonic
Continuation 2	+20 (total 70)	80.89%	Smooth, monotonic
Final Test Set Evaluation	-	81.16%	-

2.3 Training Progression Chart

The chart below shows EfficientNetB0 val accuracy at each training phase plotted against the Custom CNN best val accuracy baseline of 57.66%:



2.4 Final Confirmed Results on Test Set

The following results were confirmed on the held-out test set (5,254 images, unseen during training):

Model	Test Accuracy	Gap
Custom CNN (from scratch, 322,470 params)	52.84%	–
EfficientNetB0 (ImageNet pretrained, 4.37M params)	81.16%	+28.32 percentage points

The 28.32 percentage point gap on pooled multi-source data (ISIC 2019 + HAM10000) constitutes the core confirmed finding of the skin lesion experiment. EfficientNetB0 improved smoothly and monotonically across all 70 training epochs while the custom CNN oscillated between 11.14% and 57.66% with no stable convergence. This stability contrast directly supports the implicit domain adaptation hypothesis.

2.5 Model Weights and Artifacts

- Best EfficientNetB0 weights saved to Kaggle version history and downloaded locally (best_efficientnet_v3.keras)

- Custom CNN weights saved (best_custom_cnn.keras)
- Training logs saved as CSV for all phases (cnn_training_log.csv, efficientnet_v2_log.csv, efficientnet_v3_log.csv)
- All artifacts versioned on Kaggle with Save Version after each training block

3. Plan for Next Week (June 8-14)

- Continue EfficientNetB0 training for additional epochs when GPU quota resets (model was still improving at epoch 70, early stopping did not trigger)
- Train MobileNetV2 and ResNet50 on the same pooled dataset to complete the four-model comparison
- Run single-source baseline experiments: train each model on ISIC 2019 alone, evaluate on HAM10000 to quantify cross-device generalization gap
- Initiate PhysioNet credentialling process for CheXpert and MIMIC-CXR access (requires CITI human research training and data use agreement)
- Begin chest X-ray pipeline setup using VinDr-CXR (available on Kaggle without credentialling) as the first chest source

4. Current Progress

S.No.	Work Done	Date Completed
1	• Dataset acquisition and pipeline setup (ISIC 2019 + HAM10000, 35,019 images, 6 classes, stratified split, class weighting) • Custom CNN training (27 epochs, best val accuracy 54.58%, oscillating instability confirmed) • EfficientNetB0 Phase 1 (frozen backbone, 10 epochs, best val accuracy 65.56%)	Week 1 - 31.05.2026
2	• EfficientNetB0 Phase 2 + 2 continuation runs (70 total epochs, best val accuracy 80.89%) • Final test set evaluation: Custom CNN 52.84%, EfficientNetB0 81.16%, gap 28.32 points • All weights and logs saved and versioned on Kaggle	Week 2 - 07.06.2026

5. Issues and Risks

- **Kaggle GPU quota:** 30 hours per week free tier. EfficientNetB0 training across all phases consumed approximately 26 hours this week. Mitigation: weights saved after every run, version history used to reload between sessions, training scheduled in overnight blocks.
- **Session restart data loss:** Kaggle sessions restart when idle, wiping in-memory variables and unsaved files. Resolved by saving all weights to /kaggle/working/ with ModelCheckpoint and using Kaggle version history as permanent storage. Models are now loadable from any new session without retraining.
- **PhysioNet credentialling:** CheXpert and MIMIC-CXR require CITI human research training certification and data use agreement approval. Process will be initiated this week. Estimated approval time 3-7 days. VinDr-CXR (Vietnamese chest X-ray dataset) is available on Kaggle without credentialling and will serve as the first chest source.

6. Declaration

We declare that the work reported in this weekly report has been carried out by us as part of the Minor Specialization project in Computational Intelligence. All experiments were conducted on publicly available datasets under their respective licenses.

Abhay Sivakumar Nair

Reg. No: 220905199

Date: June 7, 2026

Shruthi Andra

Reg. No: 230905223

Date: June 7, 2026